

Android Phone Management System

- ▶ Project Presentation
- ▶ Using C++ & Data Structures

Introduction

- ▶ Simulates mobile phone operations
- ▶ Manages contacts, messages, call history
- ▶ Console-based C++ application

Objective

- ▶ Understand real-world use of data structures
- ▶ Improve C++ programming skills
- ▶ Implement efficient data handling

Modules

- ▶ Contacts → Linked List
- ▶ Messages → Queue (FIFO)
- ▶ Call History → Stack (LIFO)

System Features

- ▶ Add & view contacts
- ▶ Send & delete messages
- ▶ Make calls & view call logs
- ▶ Menu-driven interface

Algorithm

- ▶ Start program
- ▶ Display menu
- ▶ User selects option
- ▶ Perform operation using switch-case
- ▶ Repeat until exit

Methodology

- ▶ Modular design
- ▶ Separate functions for each task
- ▶ Validation using conditions
- ▶ Efficient memory management

Requirements

- ▶ Hardware: Computer, Keyboard
- ▶ Software: C++ Compiler
- ▶ OS: Windows/Linux/Mac

Output

==== ANDROID PHONE SYSTEM ====

1. Add Contact
2. View Contacts
3. Send Message
4. View Messages
5. Delete Message
6. Make Call
7. Call History
8. Delete Last Call
9. Exit

Enter choice: 1

Enter Name: VIGNESH

Enter Number: 9100598298

Contact Added!

==== ANDROID PHONE SYSTEM ====

1. Add Contact
2. View Contacts
3. Send Message
4. View Messages
5. Delete Message
6. Make Call
7. Call History
8. Delete Last Call
9. Exit

Enter choice: 2

VIGNESH - 9100598298

==== ANDROID PHONE SYSTEM ====

1. Add Contact
2. View Contacts
3. Send Message
4. View Messages
5. Delete Message
6. Make Call
7. Call History
8. Delete Last Call
9. Exit

Enter choice: 3

Enter Recipient Name: VIGNESH

Enter Message: HELLO VIGNESH

Message Sent to VIGNESH!

==== ANDROID PHONE SYSTEM ====

1. Add Contact
2. View Contacts
3. Send Message
4. View Messages
5. Delete Message
6. Make Call
7. Call History
8. Delete Last Call
9. Exit

Enter choice: 4

1. To VIGNESH: HELLO VIGNESH

==== ANDROID PHONE SYSTEM ====

1. Add Contact
2. View Contacts
3. Send Message
4. View Messages
5. Delete Message
6. Make Call
7. Call History
8. Delete Last Call
9. Exit

Enter choice: 5

Deleting: To VIGNESH - HELLO VIGNESH

==== ANDROID PHONE SYSTEM ====

1. Add Contact
2. View Contacts
3. Send Message
4. View Messages
5. Delete Message
6. Make Call
7. Call History
8. Delete Last Call
9. Exit

Enter choice: 6

Enter Contact Name: VIGNESH

Calling VIGNESH...

```
==== ANDROID PHONE SYSTEM ====
```

1. Add Contact
2. View Contacts
3. Send Message
4. View Messages
5. Delete Message
6. Make Call
7. Call History
8. Delete Last Call
9. Exit

```
Enter choice: 7  
VIGNESH
```

```
==== ANDROID PHONE SYSTEM ====
```

1. Add Contact
2. View Contacts
3. Send Message
4. View Messages
5. Delete Message
6. Make Call
7. Call History
8. Delete Last Call
9. Exit

```
Enter choice: 8  
Deleting Call: VIGNESH
```

```
==== ANDROID PHONE SYSTEM ====
```

1. Add Contact
2. View Contacts
3. Send Message
4. View Messages
5. Delete Message
6. Make Call
7. Call History
8. Delete Last Call
9. Exit

```
Enter choice: 9  
Exiting...
```

I

```
=== Code Execution Successful ===
```

Advantages

- ▶ Efficient data handling
- ▶ Faster operations
- ▶ Structured program design
- ▶ Easy to use

Conclusion

- ▶ Demonstrates practical use of data structures
- ▶ Improves problem-solving skills
- ▶ Useful academic project

Future Enhancements

- ▶ Add GUI interface
- ▶ Database storage
- ▶ Search & edit contacts
- ▶ Security features